RESEARCH SHOWCASE / AI SECURITY REPAIR

The Tests Passed. The Grader Said No.

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RESEARCH QUESTION

Can we judge AI decisions when the checker misleads?

BACKGROUND

AI revises code using feedback [1, 2]. We checked the feedback itself.

METHODS

40 sessions · 1 local model
5 Python security task types
Compare final tests with verdicts.

RESULTS / 01 — THE HIDDEN RULE

9 / 20

Passed the tests. Rejected anyway.



9 passed tests
11 failed tests

Our hidden rule rejected all 20.

The AI could not see the rule. Test pass ≠ proof of security.

RESULTS / 02 — STOPPING

2 min 8 sec

Median: stop acknowledgement → run end.
13 repairable post-PASS sessions.
A 14th timed out (excluded).

0 further logged repairs

The old controller waited for another final signal.

CONCLUSION

Check the grader.

NEXT STEPS

Independent task review → broader paired runs.
We improved logging and tested immediate stopping.

REFERENCES [1] Yao et al., ReAct (2023), arXiv:2210.03629. [2] Shinn et al., Reflexion (2023), arXiv:2303.11366. [3] github.com/Imranqsl212/AiResearch